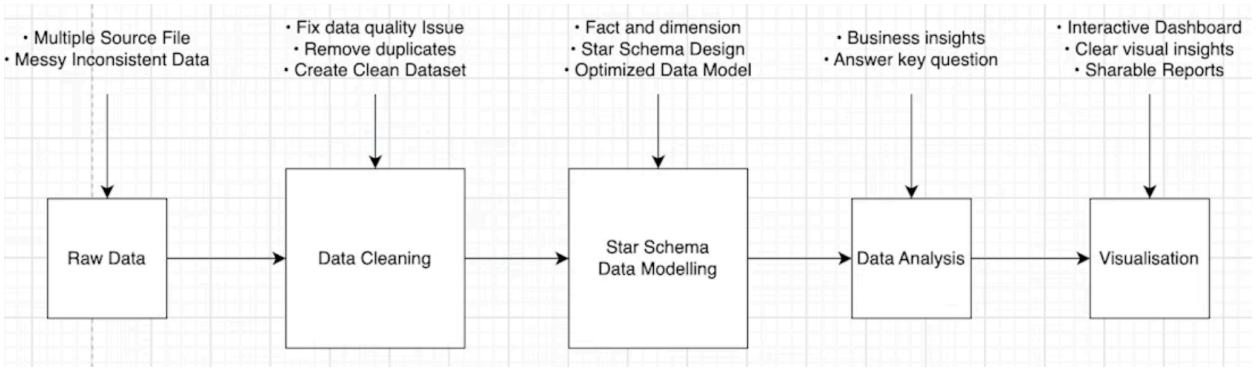
AI-Powered Data Analytics

Prompt Engineering, Exploratory Data Analysis (EDA), Data Cleaning, Data Modeling, and Dashboarding.

Tools : *Cursor AI: Grok 4.6 Medium*

Key Competencies: Data quality assessment, data transformation, analytical modeling, KPI identification, dashboard development, and business insight generation.

Work-Flow :



Prompt *(data inspection)* **:**

> I have provided 4 csv data files , read them DO NOT write code or perform analysis yet.
> I am working on a project. The aim of the project is to get Clean data,Analyse it , build interactive dashboard & get insights from raw data & suggest business strategies.
> For now just give me basic overview of data like stats summary,data shape , data description ,inconsistencies or invalid data (if any) and any other data quality issue you find
>
> IMPORTANT INSTRUCTIONS :
> DO NOT write code
> DO NOT modify data provided
> DO NOT perform data cleaning
>
> Purely based on data provided suggest next steps

Response :

I'll locate the four CSV files and inspect them directly so I can give you a quality overview without writing code, changing the data, or running analysis.
The four files are `orders_raw.csv`, `customers_raw.csv`, `products_raw.csv`, and `regions_raw.csv`. Next I'll inspect their sizes, columns, and a wide sample of rows.
The smaller tables look inspectable by hand; I'll scan the larger ones for missing values, duplicates, invalid ranges, and broken keys without saving any scripts or changing files.
Customers and orders look like they have duplicate IDs at the end. I'll tally uniqueness, ranges, blanks, and key mismatches across all four files next.
Sourabh, this is a synthetic India retail sales dataset: 4 linked tables covering 2024–2025. Nothing was changed, and no cleaning or analysis was done.

How the files connect

`orders` is the fact table. It should join to:

- `customers` on `customer_id`
- `products` on `product_id`
- `regions` on `region_id`

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1. `customers_raw.csv`

Shape: 2,010 rows × 6 columns (header + 2,010 records). 2,000 unique customers; 10 exact duplicate rows appended at the end.

Columns: `customer_id`, `customer_name`, `gender`, `age`, `customer_segment`, `signup_date`

Description: One row per customer. IDs are `C00001`–`C02000`. Names are placeholders (`Customer 1`, `Customer 2`, ...).

Unique-customer snapshot (after ignoring the 10 copies):

Field	What it looks like
Gender	Male 1,043 · Female 910 · Other 47

Segment	Consumer 1,271 · Corporate 444 · Small Business 285
Age	18–65, mean ~41.2; all integers; no blanks
Signup	2024-01-01 to 2025-12-30; format `YYYY-MM-DD`; no blanks

Quality issues: duplicate IDs only. No missing values, no age outliers, no extra whitespace.

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2. `products_raw.csv`

Shape: 100 rows × 6 columns. All `product_id` values are unique (`P0001`–`P0100`).

Columns: `product_id`, `product_name`, `category`, `sub_category`, `cost`, `selling_price`

Description: Catalog of 100 SKUs. Names are generic (`Stationery Product 1`). Cost 14.82–897.03; selling price 20.60–1,457.27. Catalog list margin is about 12–42% (mean ~26%). The cost is never above the selling price. No blanks.

Quality issues (this table is the messiest text-wise):

- Inconsistent category labels: `Electronics` vs `electronics`; `Office Supplies` vs `OFFICE SUPPLIES`; `Furniture` vs `Furniture ` (trailing space). That is 8 raw labels for what should be ~5 categories.
- Category does not match subcategory on at least 12 products, for example:
    - Stationery / Paper tagged as Furniture or electronics
    - Men / Women tagged as OFFICE SUPPLIES or electronics
    - Mobiles tagged as Furniture or OFFICE SUPPLIES
    - Appliances tagged as electronics
    - Tables tagged as OFFICE SUPPLIES
- Product name and subcategory do agree; the category field is the broken one.

If you later group sales by category without standardizing this, Electronics / Furniture / Office Supplies will be wrong.

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3. `regions_raw.csv`

Shape: 20 rows × 4 columns. All IDs unique (`R001`–`R020`). No blanks, no duplicates.

Columns: `region_id`, `state`, `city`, `region`

Zones: West 4 · North 5 · South 5 · East 5 · Central 1 (only Madhya Pradesh)

Quality / realism issue: every city is a placeholder: `Maharashtra City`, `Delhi City`, etc. (`city` = `state` + `" City"` for all 20 rows). Fine for a class project, not usable as real city-level geography.

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4. `orders_raw.csv`

Shape: 25,050 rows × 10 columns. 25,000 unique `order_id`s; 50 exact duplicate rows at the end of the file.

Columns: `order_id`, `order_date`, `customer_id`, `product_id`, `region_id`, `quantity`, `discount`, `sales_amount`, `cost_amount`, `profit`

Description: Line-level orders (one product per order). IDs `O000001`–`O025000`. Time window 2024-01-01 to 2025-12-31, split almost evenly (2024 ≈ 12,498; 2025 ≈ 12,542). Every valid timestamp is midnight (`00:00:00`), so there is no time-of-day.

Numeric snapshot (raw file, including dups):

Field	Range / notes
Quantity	−5 to 5; mostly 1–2
Discount	only 0, 0.05, 0.10, 0.15, 0.20
Sales	−2,900.28 to 7,286.35
Cost	14.82 to 4,485.15 (never negative)
Profit	−1,110.80 to 3,014.30; 586 negative-profit rows

For positive quantity, sales matches `selling_price × qty × (1 − discount)`, and profit matches `sales − cost`. Negative profit on valid qty (571 rows) is mostly heavy discount vs thin margin — a business finding, not a parse error.

Invalid / inconsistent data:

Issue	Count	Detail
Duplicate order rows	50	identical copies of existing IDs, last 50 rows
Blank `customer_id`	60	no unknown IDs otherwise
Blank `product_id`	35	no unknown IDs otherwise
Invalid `order_date`	10	literal value `not_a_date`
Negative quantity	15	sales go negative; cost stays positive
Profit formula broken	those same 15	`profit ≠ sales − cost` because cost did not flip sign
Orders before customer signup	~12,535 unique orders (~50%)	customer “lifetime” dates are not trustworthy
Missing keys vs bad dates vs neg qty	no overlap	three separate defect types

No orphan region IDs. Every customer and every product appears in orders at least once (except the blank-key rows).

Cross-table / overall quality

This looks intentionally dirty teaching data, not a production extract.

What is solid

- Key formats are consistent (`C#####`, `P####`, `R###`, `O######`)
- No extra columns, no encoding junk, no mixed delimiters
- Product prices and (positive) order amounts are internally consistent

- Regions and customer attributes are complete

What will break cleaning, joins, and a dashboard if ignored

1. Duplicate rows in customers (10) and orders (50)
2. Missing foreign keys on orders (60 + 35)
3. Ten unusable dates (`not_a_date`)
4. 15 nonsense quantities (returns? data error? cost sign is wrong either way)
5. Product category casing, whitespace, and wrong category-subcategory pairs
6. Placeholder names (customers, products, cities)
7. Signup vs order-date conflict on about half of orders
8. Central zone has only one state, so region comparisons will look unbalanced

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Suggested next steps (plan only — no cleaning yet)

Do this in order before analysis or the dashboard:

1. Lock a cleaning rulebook (document, not code yet): how you will treat duplicates, blanks, `not_a_date`, negative qty, and category labels. Decide whether negative qty is a return or an error.
2. Define grain: one row = one order line. Unique business keys: `customer_id`, `product_id`, `region_id`, `order_id`.
3. Clean in a copy of the data, never the raw CSVs: drop exact duplicates → fix/flag invalid dates → handle missing FKs (exclude from joins or keep in an “unknown” bucket) → fix quantity/cost sign → standardize category text → realign category to subcategory using a mapping table.
4. Rebuild integrity checks: every order customer/product/region exists; `sales ≈ price × qty × (1 − discount)`; `cost ≈ catalog_cost × qty`; `profit = sales − cost`; `order_date ≥ signup_date` (or flag exceptions).
5. Add analysis columns after cleaning: year/month, zone, standardized category, discount band, customer age band, margin %, new vs repeat customer (only if you resolve the signup-date problem).
6. Then analyse: sales/profit by time, category, zone, segment, gender, discount. Look at unprofitable discounting, SKU mix, regional gaps (Central vs others), and corporate vs consumer value.
7. Dashboard last: KPI header (orders, sales, profit, AOV, margin) + filters (date, region, category, segment) + trend, category/region bars, discount vs margin, customer mix. Call out data caveats (synthetic names/cities; flagged rows).
8. Strategy ideas only after that: e.g. cap discounts that wipe margin, fix mis-tagged SKUs before assortment decisions, don’t over-interpret city or “new customer” metrics until those fields are repaired.

When you want to move on, the first implementation step is the cleaning rulebook + a cleaned copy of these four files — still leaving `*_raw.csv` untouched.

Prompt *(data cleaning & preprocessing)* :

> Now clean data , handle missing , duplicates values, invalid dates, create flags if necessary, generate new files for cleaned data

Response :

I'll inspect the project layout, then write cleaned copies of the four files (raw files stay untouched), with flags for missing keys, bad dates, returns, and category fixes.
The four `*_raw.csv` files were left unchanged. Cleaned copies are in `data/`:

File	Rows	Notes
`data/customers_clean.csv`	2,001	2,000 unique customers + `C_UNKNOWN`
`data/products_clean.csv`	101	100 SKUs + `P_UNKNOWN`
`data/regions_clean.csv`	20	same 20 regions
`data/orders_clean.csv`	25,000	one row per unique `order_id`

What was cleaned

Duplicates

Dropped 10 exact duplicate customer rows and 50 exact duplicate order rows (kept the first copy).

Missing keys

60 blank `customer_id` values → `C_UNKNOWN`.

35 blank `product_id` values → `P_UNKNOWN`.

Matching placeholder rows were added so joins still work. Order amounts were not dropped.

Invalid dates

10 `not_a_date` values were cleared (`order_date` left blank). Valid timestamps were standardized to `YYYY-MM-DD`.

Negative quantity

15 rows treated as returns (`is_return=1`). `cost_amount` was given the same sign as quantity, then `profit = sales - cost` was recomputed. Sales formula still matches the catalog.

Product categories

Casing/whitespace standardized. Category was aligned to subcategory (subcategory was the reliable field). 12 SKUs were remapped (e.g. Stationery no longer under Furniture).

Whitespace

All text fields trimmed.

Flags added (use these in analysis / dashboard filters)

Orders

Flag	Count = 1	Meaning
`missing_customer_id`	60	original customer key was blank
`missing_product_id`	35	original product key was blank
`invalid_order_date`	10	date was `not_a_date`
`is_return`	15	quantity < 0
`cost_sign_corrected`	15	cost sign was fixed on those returns
`order_before_signup`	12,535	order date is before that customer's signup

Products: `category_original`, `category` (final), `category_corrected` (12 rows).
Regions: `placeholder_city` (all 20 — cities are still `State City` placeholders; they were flagged, not invented).

Integrity after cleaning

- All IDs unique; every order key exists in the dimension tables
- No leftover `not_a_date`
- `sales = selling_price × qty × (1 − discount)`
- `cost = catalog cost × qty` (including negative qty)
- `profit = sales − cost`
- Category vs subcategory pairs are consistent

How to use this next

For KPIs and the dashboard, start from the `*_clean.csv` files.

Exclude or slice with flags when the metric requires a complete key or date, for example:

- time trends → drop `invalid_order_date = 1`
- customer segment / gender → drop `missing_customer_id = 1`
- category mix → drop `missing_product_id = 1`
- new vs repeat customers → treat `order_before_signup = 1` as unreliable (~half of orders)

Prompt *(data engineering)* :

Create a date table using valid dates from the cleaned_orders table. Include date_key,date,day,month,month name,quarter & year.Create one row per unique data & save it as dimension table.

Response :

The date dimension is saved as `data/dim_date.csv`. Raw and cleaned fact files were not changed.

Shape: 731 rows × 7 columns — one row per unique valid `order_date` from `orders_clean.csv`. The 10 blank dates (`invalid_order_date = 1`) were excluded.

Range: 2024-01-01 to 2025-12-31 (every calendar day in that window appears in orders).

Column	Example	Meaning
`date_key`	`20240101`	integer-style key `YYYYMMDD`
`date`	`2024-01-01`	join key to `orders_clean.order_date`
`day`	`1`	day of month
`month`	`1`	month number
`month_name`	`January`	full month name
`quarter`	`1`	1–4
`year`	`2024`	year

Join for analysis: `orders_clean.order_date = dim_date.date`.

Prompt *(data modeling) :*

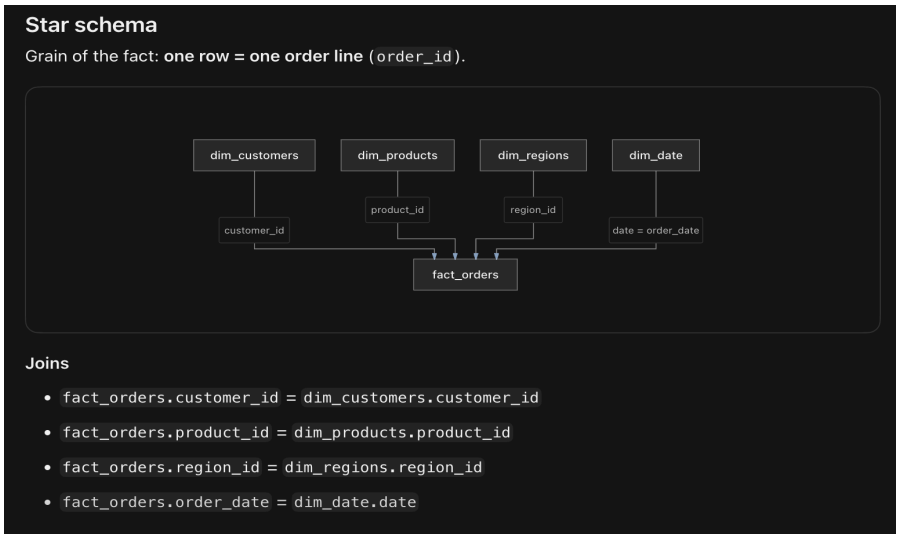
Is our data/tables in the star schema ? If yes, show structure. Also rename dimension tables with prefix 'dim' & 'fact' for fact table (only on cleaned data)

Response : The cleaned data is a star schema: one fact table in the middle, four dimensions around it. Raw files were not renamed.

Renames (cleaned only)

Old name	New name	Role
`orders_clean.csv`	`fact_orders.csv`	fact
`customers_clean.csv`	`dim_customers.csv`	dimension
`products_clean.csv`	`dim_products.csv`	dimension
`regions_clean.csv`	`dim_regions.csv`	dimension
`dim_date.csv`	`dim_date.csv`	dimension (already prefixed)

Raw files stay `*_raw.csv`.



Prompt *(data validation & testing) :*

Top 10 products by revenue ?

Response :

Top 10 products by net revenue (`SUM(sales_amount)` from `fact_orders`, including returns; 35 unknown-product rows excluded).

Rank	Product	Category	Revenue	Share of catalog	Profit
1	Kids Product 98 (`P0098`)	Clothing	829,113	2.81%	319,536
2	Tables Product 97 (`P0097`)	Furniture	728,151	2.47%	211,154
3	Storage Product 83 (`P0083`)	Furniture	699,203	2.37%	184,758
4	Appliances Product 99 (`P0099`)	Home & Kitchen	694,713	2.36%	228,439
5	Appliances Product 50 (`P0050`)	Home & Kitchen	668,335	2.27%	202,509

6	Cookware Product 59 (`P0059`)	Home & Kitchen	657,901	2.23%	196,912
7	Kids Product 80 (`P0080`)	Clothing	641,467	2.18%	161,426
8	Women Product 67 (`P0067`)	Clothing	634,806	2.15%	144,817
9	Chairs Product 12 (`P0012`)	Furniture	568,745	1.93%	196,533
10	Paper Product 94 (`P0094`)	Office Supplies	523,043	1.78%	78,887

These 10 SKUs are 22.6% of known-product revenue (₹29.46M total). Paper Product 94 ranks 10th on sales but is much weaker on profit than the others.

Prompt *(interactive dashboard) :*

Create an interactive powerBI dashboard showing revenue, profit, orders, customers, monthly sales trend, revenue by category, profit by region and top products. Add filters for year, region and category

Response :

The workspace already has a star-schema dataset. I'll inspect the tables, then build an interactive Power BI–style dashboard with the KPIs and slicers you asked for.
I'll join the star schema into a compact dataset, then build a Power BI–style HTML dashboard with live slicers.

Building the dashboard UI with KPI cards, slicers, and the four chart visuals.

The interactive dashboard is live at http://127.0.0.1:8765/dashboard/. Power BI Desktop is not available on macOS, so this is a Power BI–style report on your star schema (`fact_orders`, `dim_products`, `dim_regions`).

Slicers (left): Year, Region, Category — all visuals update together.

KPI cards: Revenue, Profit (with margin), Orders, Customers.

Visuals:

- Monthly sales trend
- Revenue by category
- Profit by region
- Top 10 products by revenue

Unfiltered totals (returns and invalid dates excluded): ₹2.95 Cr revenue, ₹67.62 L profit (23.9% margin), 24,975 orders, 2,001 customers.

Verified in the browser: 2025-only dropped orders to 12,513; 2025 + Clothing only showed ₹35.58 L and clothing products; unchecking South removed that region from the profit chart.

Open the page above, or from the project root:

```
python3 -m http.server 8765
```

Then go to `http://127.0.0.1:8765/dashboard/`. Refresh `data.js` after CSV changes with `python3 scripts/generate_dashboard_data.py`.

